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Truck towing system

Abstract

A truck towing system is described. The truck towing system provides for a first truck to tow a second truck in a back to front relationship, i.e., the front portion of the second truck (the towed truck) is connected or attached to a rear portion of the first truck (the towing truck). The truck towing system includes an upper member to connect with a first truck. A middle member joins the upper member and a lower member in a fixed engagement. The lower member supports a front portion of a second truck. A beam member is engaged to the lower member. The beam member connects to a rear portion of the second truck.

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Background/Summary

(1) This application claims priority to U.S. Provisional Patent Application No. 63/145,591 filed Feb. 4, 2021.

FIELD OF INVENTION

(1) The present invention relates to a truck towing system.

BACKGROUND OF INVENTION

(2) Previous attempts of towing or transporting semi-trucks have used a method called “decking,” in which a first truck tows a second truck with an axle assembly of the second truck positioned over a portion or frame of the first truck. The method does not work with many modern trucks that have an increased height, since a top of the second truck will be too high off of the ground resulting in problems passing through tunnels or under over-passes and possible violation of laws or regulations directed to maximum vehicle height.

(3) Other attempts in towing semi-trucks use a method in which the truck that is being towed is

towed in reverse. This method is fuel inefficient since the towed truck is moving in a direction counter to its aerodynamic design. Further, towing in reverse may damage the suspension or the air fairings of the towed truck.

(4) U.S. Pat. Nos. 8,371,599, 8,876,139, and 9,636,957, all assigned to Truckmovers.COM, Inc. of Kansas City, Missouri are hereby incorporated by reference. These patents improve on the previous attempts in the towing of semi-trucks.

SUMMARY OF INVENTION

(5) A truck towing system is herein described. The truck towing system provides for a first truck to tow a second truck in a back to front relationship, i.e., the front portion of the second truck (the towed truck) is connected or attached to a rear portion of the first truck (the towing truck). This allows the first truck to tow the second truck in a fuel-efficient manner, as trucks are not generally designed to be aerodynamic when driven in or pulled in a reverse direction.

(6) The truck towing system is convertible for use between towing trucks with a saddle mounts, fifth wheel mounts, or other hitches and mounts. The towing system includes one or more connection assemblies that are configured to engage with the saddle mounts, fifth wheel mounts, or other hitches and mounts of the towing truck. For example, a first connection assembly may be removed from the towing system and replaced with a second connection assembly. The one or more connection assemblies are configured to operate with a specific mount of the towing truck, such as saddle mounts, fifth wheel mounts, or other hitches and mounts. Thus, a single truck towing system may be provided with different connection assemblies to match a specific mount of the towing truck. A front portion of the connection assembly removably engages or connects with the towing truck, while a rear portion of the connection assembly removably engages or connects with the remainder of the truck towing system.

(7) The truck towing system also allows for the towed truck to be maintained at a height where a top surface or a roof of the towed truck is maintained below approximately 13 feet, 6 inches, which is a typical maximum height for safe clearance on most highways and interstates. Many of today's semi-trucks are built with cabins or condos above and behind the driver's seat in the cab. These areas may contain a bed or other storage area. These additions to the cab create extra height that must be accommodated by the towing equipment.

(8) The truck towing system allows a single driver to move two or more trucks to a destination. This reduces labor and travel costs since two drivers are not required.

(9) The truck towing system further provide for the towed truck to have its rear wheels on the ground, while its front wheels are lifted. This provides the driver of the first truck with better control over the second truck. This also reduces the likelihood of the towed trucked to start "snaking," which may result in a loss of control.

(10) The truck towing system further is adjustable to accommodate different sized semi-trucks. The system may extend or retract depending on the length of the towed vehicle.

(11) In another aspect, a truck towing system is provided. The truck towing system includes an upper member to removably connect with a first truck, a middle member, and a lower member. The middle member joins the upper member and the lower member. The lower member supports a front portion of a second truck. A beam member is engaged to the lower member to pass through a front opening and a rear opening of the lower member. The lower member defines an interior to receive the beam member. The beam member connects to a rear portion of the second truck. The lower member includes one or more sets of openings that are spaced along the length of the lower member. The beam member includes one or more sets of pin openings that are spaced along the length of the beam member. A locking pin passes through one of the sets of the openings in the lower member and one of the sets of the pin openings in the beam member to lock the beam member and the lower member together.

(12) In another aspect, a method of connecting a truck towing system to a first truck and a second truck is provided. The method includes providing a towing system having an upper member that

removably connects with the first truck, a middle member, and a lower member. The middle member joins the upper member and the lower member. The lower member supports a front portion of a second truck. A beam member is engaged to the lower member. The beam member connects to a rear portion of the second truck. The method further includes extending or retracting the beam member relative to the lower member. The method further includes locking the beam member and the lower member together. The method further includes positioning the second truck over the lower member and the beam member and locking a front axle assembly of the second truck to the lower member. The method further includes locking a rear frame of the second truck to the beam member and lifting the upper member. The method further includes attaching the first truck to the upper member of the towing system.

(13) In another aspect, a method of storing a truck towing system is provided. The method includes providing a towing system having an upper member to removably connect with a first truck, a middle member, and a lower member. The middle member joins the upper member and the lower member. The lower member supports a front portion of a second truck. A beam member is engaged to the lower member. The beam member connects to a rear portion of the second truck. The method further includes retracting the beam member relative to the lower member through an interior of the lower member and through a front opening of the lower member.

(14) In another aspect, a towing system that is convertible for use between towing trucks with a saddle mounts, fifth wheel mounts, or other hitches and mounts is described. The towing system includes one or more connection assemblies that are configured to engage with the saddle mounts, fifth wheel mounts, or other hitches and mounts of the towing truck. A front portion of the connection assemblies removably engages or connects with the towing truck. A rear portion of the connection assemblies removably engages or connects with an upper member of the towing system. The different connection assemblies are easily removable and may include front portions with kingpin or frame connection assemblies. The towed truck is engaged to the lower member of the towing system.

(15) In another aspect, a truck towing system includes an upper member. A connection assembly is configured to removably engage or connect with a first truck. The connection assembly is configured to removably engage or connect with the upper member. A lower member joins the upper member in a fixed engagement. The lower member supports a front portion of a second truck. A beam member is engaged to the lower member. The beam member slides in and out of an interior of the lower member. The beam member is configured to connect to a rear portion of the second truck.

(16) In another aspect, a truck towing system for use with a saddle mount or a kingpin mount is described. The truck towing system includes an upper member. A connection assembly is configured to removably engage or connect with a towing truck having the saddle mount or the kingpin mount. The connection assembly is a frame connection assembly or a kingpin connection assembly. The connection assembly is configured to removably engage or connect with the upper member. A lower member joins the upper member in a fixed engagement. The lower member is configured to support a front portion of a towed truck. A beam member is engaged to the lower member. The beam member is configured to connect to a rear portion of the towed truck.

(17) In another aspect, a truck towing system includes an upper member configured to removably engage or connect with a first truck. A lower member joins the upper member in a fixed engagement. The lower member is configured to support a front portion of a second truck. A beam member is engaged to the lower member. The beam member slides in and out of an interior of the lower member. The beam member is configured to connect to a rear portion of the second truck. The beam member includes a first transport opening and a second transport opening. The lower member includes a third transport opening. The first transport opening, the second transport opening, and the third transport opening provide for access or entry points for a forklift, skid-steer, or other moving apparatus with a fork or arm members to engage with the towing system in order

to lift and move the towing system. In certain aspects, the beam is withdrawn or retracted such that the first transport opening is aligned with the third transport opening in the lower member. In certain aspects, the second transport opening is outside of the lower member.

(18) In another aspect, a method of converting a truck towing system for use with a saddle mount or a kingpin mount is described. The method includes providing an upper member and a lower member. The lower member joins the upper member in a fixed engagement. The lower member is configured to support a front portion of a towed truck. The method includes providing a beam member. The beam member is configured to engage to the lower member. The beam member slides in and out of an interior of the lower member. The beam member is configured to connect to a rear portion of the towed truck. The method includes engaging or connecting a frame connection assembly or a kingpin connection assembly with the upper member. The frame connection assembly and the kingpin connection assembly are configured to removably engage or connect with the saddle mount or the kingpin mount of a towing truck.

(19) In another aspect, a method of configuring a truck towing system for use with either a saddle mount or a kingpin mount is described. The method includes providing an upper member and a lower member. The lower member joins the upper member in a fixed engagement. The lower member is configured to support a front portion of a towed truck. The method includes providing a beam member. The beam member is configured to engage to the lower member. The beam member slides in and out of an interior of the lower member. The beam member is configured to connect to a rear portion of the towed truck. The method includes selecting a frame connection assembly or a kingpin connection assembly. The frame connection assembly is configured to engage or connect with the saddle mount, and the kingpin connection assembly is configured to engage or connect with the kingpin mount. The method includes engaging or connecting the selected connection assembly with the upper member.

(20) In another aspect, a method of moving a truck towing system is described. The method includes providing the truck towing system. The truck towing system including an upper member configured to removably engage or connect with a first truck. A lower member joins the upper member in a fixed engagement. The lower member is configured to support a front portion of a second truck. A beam member is engaged to the lower member. The beam member slides in and out of an interior of the lower member. The beam member is configured to connect to a rear portion of the second truck. The beam member includes a first transport opening and a second transport opening. The lower member includes a third transport opening. The method includes moving the beam member to align the first transport opening with the third transport opening. The method includes engaging the first transport opening, the second transport opening, and the third transport opening with a forklift, skid-steer, or other moving apparatus. In certain aspects, a first fork or arm and a second fork or arms generally simultaneously pass through or into the first transport opening, the second transport opening, and the third transport opening. In certain aspects, the first fork or arm of the moving apparatus passes through or into a combination of the first transport opening and the third transport opening, while the second fork or arm of the same moving apparatus passes through or into the second transport opening.

(21) In another aspect, a method of configuring a truck towing system for use with either a saddle mount or a kingpin mount is described. The method includes providing an upper member and a lower member, wherein the lower member joins the upper member in a fixed engagement, the lower member configured to support a front portion of a towed truck. The method includes providing a beam member, the beam member fixedly engaged to the lower member, and the beam member configured to connect to a rear portion of the towed truck. The method includes selecting a frame connection assembly or a kingpin connection assembly, the frame connection assembly configured to engage or connect with the saddle mount, and the kingpin connection assembly configured to engage or connect with the kingpin mount. The method includes engaging or connecting the selected connection assembly with the upper member.

(22) In another aspect, a method of converting a truck towing system for use with a saddle mount or a kingpin mount is described. The method includes providing an upper member and a lower member. The lower member joins the upper member in a fixed engagement. The lower member is configured to support a front portion of a towed truck. The method includes providing a beam member. The beam member is engaged to the lower member. The beam member is configured to connect to a rear portion of the towed truck. The method includes determining whether a towing truck has a saddle mount or a kingpin mount. The method includes engaging or connecting a frame connection assembly or a kingpin connection assembly with the upper member. The frame connection assembly is configured to removably engage or connect with the saddle mount of the towing truck. The kingpin connection assembly is configured to removably engage or connect with the kingpin mount of the towing truck.

(23) In another aspect, a method of using of a truck towing system is described. The method includes providing an upper member and a lower member, wherein the lower member joins the upper member in a fixed engagement, and the lower member is configured to support a front portion of a towed truck. The method includes providing a beam member. The beam member is engaged to the lower member. The beam member is configured to connect to a rear portion of the towed truck. The method includes providing a frame connection assembly and a kingpin connection assembly. The method includes selecting the frame connection assembly or the kingpin connection assembly. The frame connection assembly is configured to engage a saddle mount of a towing truck. The kingpin connection assembly is configured to engage a kingpin mount of the towing truck. The method includes engaging the selected connection assembly with the upper member. The method includes engaging the selected connection assembly with the mount of the towing truck.

(24) In another aspect, a kingpin connection assembly for a towing system is described. The kingpin connection assembly includes a tube. A coupler plate is mounted to a front portion of the tube. A kingpin is mounted to the coupler plate. A rear portion of the tube is configured to removably engage or connect with an upper member of a towing system. The rear portion of the tube includes one or more sets of openings passing through lateral sides of the tube. Removable fasteners are configured to hold the rear portion of the tube in a front portion of the upper member. The front portion of the upper member may include one or more sets of openings passing through lateral sides of the front portion, and the removable fasteners pass through the openings of the lateral sides of the tube and the openings of the lateral sides of the front portion.

(25) In another aspect, a frame connection assembly for a towing system is described. The frame connection assembly includes a tube. A hitch tube is mounted to a lower surface of the tube generally perpendicular to the tube. A rear portion of the tube is configured to removably engage or connect with an upper member of a towing system. The rear portion of the tube includes one or more sets of openings passing through lateral sides of the tube. Removable fasteners are configured to hold the rear portion of the tube in a front portion of the upper member. The front portion of the upper member may include one or more sets of openings passing through lateral sides of the front portion, and the removable fasteners pass through the openings of the lateral sides of the tube and the openings of the lateral sides of the front portion.

Description

DESCRIPTION OF FIGURES

(1) FIG. 1 is a perspective view of the truck towing system with the interchangeable frame connection assembly and kingpin connection assembly.

(2) FIG. 2 is a perspective view of the truck towing system with the frame connection assembly.

(3) FIG. 3 is a top view of the truck towing system with the frame connection assembly.

- (4) FIG. 4 is a side view of the truck towing system with the frame connection assembly.
- (5) FIG. 5 is a side view of the truck towing system with the frame connection assembly and the beam member in the collapsed configuration.
- (6) FIG. 6 is a view of the first truck towing the second truck with the truck towing system with the frame connection assembly.
- (7) FIG. 7 is a side view of the connection at the rear portion of the second truck.
- (8) FIG. 8 is a side sectional view of the connection at the rear portion of the second truck.
- (9) FIG. 9 is a front sectional view of the connection at the rear portion of the second truck.
- (10) FIG. 10 is a side view of the connection to the frame connection assembly at the rear portion of the first truck.
- (11) FIG. 11 is a perspective view of the frame connection assembly.
- (12) FIG. 12 is a top view of the frame connection assembly.
- (13) FIG. 13 is a bottom view of the frame connection assembly.
- (14) FIG. 14 is a side view of the frame connection assembly.
- (15) FIG. 15 is a front view of the frame connection assembly.
- (16) FIG. 16 is a rear view of the frame connection assembly.
- (17) FIG. 17 is a perspective view of the kingpin connection assembly.
- (18) FIG. 18 is a lower perspective view of the kingpin connection assembly.
- (19) FIG. 19 is a top view of the kingpin connection assembly.
- (20) FIG. 20 is a bottom view of the kingpin connection assembly.
- (21) FIG. 21 is a side view of the kingpin connection assembly.
- (22) FIG. 22 is a front view of the kingpin connection assembly.
- (23) FIG. 23 is a rear view of the kingpin connection assembly.
- (24) FIG. 24 is a top view of the towing system with the frame connection assembly installed.
- (25) FIG. 25 is a side view of the towing system with the frame connection assembly installed.
- (26) FIG. 26 is a top view of the towing system with the kingpin connection assembly installed.
- (27) FIG. 27 is a side view of the towing system with the kingpin connection assembly installed.
- (28) FIG. 28 is a side view of the towing system with the beam member removed.
- (29) FIG. 29 is a side view of the beam member.
- (30) FIG. 30 is a side view of the towing system with the kingpin connection assembly installed and the beam member collapsed.
- (31) FIG. 31 is a side view of the towing system with the kingpin connection assembly installed and the beam member extended.

DETAILED DESCRIPTION OF INVENTION

- (32) A truck towing system **40** will now be described with reference to the FIGURES. A truck towing system **40** is shown in FIG. 1 with a kingpin connection assembly **500** and a frame connection assembly **600**. The kingpin connection assembly **500** and the frame connection assembly **600** may be removed from and interchangeably used with the truck towing system **40** depending upon the towing set-up needed by a user.
- (33) The towing system **40** includes an upper member **100** that is configured to removably connect or engage with the kingpin connection assembly **500** and the frame connection assembly **600**. The frame connection assembly **600** connects to a saddle mount **15** of a first truck **10**, while the kingpin connection assembly **500** connect to a fifth wheel mount (not shown) of the first truck **10** or a different truck. Thus, the truck towing system **40** may be used with towing trucks having either a saddle mount or a fifth wheel mount by switching between the kingpin connection assembly **500** and the frame connection assembly **600**.
- (34) With reference to FIG. 6, the system **40** connects to a rear portion of the first truck **10** and to a front portion of a second truck **20**. The first truck **10** is the towing truck, while the second truck **20** is the towed truck. The second truck **20** is towed in a forward direction. A third truck may also be towed by the second truck **20** in a similar manner.

(35) The first truck **10** includes the saddle mount **15**, which receives the towing system **40** in a fixed and locking connection. A front axle assembly **23** of the second truck **20** rests on the towing system **40**. A rear frame **26** of the second truck **20** is locked, by chains or otherwise clamping, to the system **40**. Rear wheels tandem **29** of the second truck **20** are in contact with the ground and roll as the first truck **10** pulls or tows the second truck **20**. Accordingly, the second truck **20** is being pulled in the direction in which the second truck **20** was aerodynamically designed. As such, pulling the second truck **20** in the forward direction results in fuel economy savings as the first truck **10** is performing reduced work verses towing the second truck **20** in a backward manner. In this aspect, a front portion **118** of the upper member **100** engages with or receives either the kingpin connection assembly **500**, the frame connection assembly **600**, or another style of connection assembly.

(36) The towing system **40** includes the upper member **100**, a middle member **200**, and a lower member **300**. Each of the members **100**, **200**, and **300** include a generally linear length of a rigid and strong metal or metal alloy. Each of the members **100**, **200**, and **300** may be formed from solid, hollow or partially hollow sections of square tubing or stock. The members **100**, **200**, and **300** may be made from low to medium strength carbon steel. The upper member **100** is connected to the middle member **200**, and the middle member **200** is connected to the lower member **300**. The upper member **100**, the middle member **200**, and the lower member **300** may be fixedly connected to each other, or the members **100**, **200** and **300** may be an integral structure. In the aspect shown in the FIGURES, the upper member **100** is welded to the middle member **200**, and the middle member **200** is welded to the lower member **300**. Typically, none of the upper member **100**, the middle member **200**, and the lower member **300** move or pivot relative to each other.

(37) The upper member **100** is orientated in a range of approximately parallel to the lower member **300** or slightly angling toward a plane of the lower member **300** in an amount of approximately 2 degrees to 5 degrees in the direction attaching toward the first truck **10**. The upper member **100** is separated from the lower member **300** by the middle member **200**. The middle member **200** joins an end of the upper member **100** with an end of the lower member **300**.

(38) With reference to FIG. 2, the upper member **100** includes the front portion **118** and a rear portion **160**. The kingpin connection assembly **500** and the frame connection assembly **600** removably connect or removably engage to the front portion **118**. The frame connection assembly **600** removably connects or engages to the saddle mount **15** of the first truck **10**. The kingpin connection assembly **500** removably connects or engages to a fifth wheel mount (not shown) of the first truck **10**.

(39) The rear portion **160** of the upper member **100** includes a rear end **162** that defines rear surfaces **164**. The rear end **162** and the rear surfaces **164** are fixedly engaged or connected by welding or other fasteners to the middle member **200**. The upper member **100** further includes side surfaces **166**. Connecting plates **180** connect the side surfaces **166** of the upper member **100** with side surfaces **222** of the middle member **200**. The connecting plates **180** strengthen or reinforce the connection or bond between the upper member **100** and the middle member **200**.

(40) The middle member **200** includes an upper portion **220** and a lower portion **260**. The upper portion **220** is fixed or connected to the rear portion **160** of the upper member **100**. The middle member **200** is fixed or connected to the upper member **100** at an angle of approximately 100 degrees to approximately 150 degrees. This range of angles provides for the lower member **300** to be at the proper position for connecting to the second truck **20**. With reference to FIGS. 2 and 3, the upper portion **220** includes an upper surface **210**, the side surfaces **222**, and the rear surface **230**. The upper surface **210** receives angled support plates **280** in a fixed or permanent connection that further connects the middle member **200** with the lower member **300**.

(41) The lower member **300** will now be described with reference to FIG. 2. The lower portion **260** of the middle member **200** is fixed or connected to a front portion **320** of the lower member **300**. The middle member **200** is fixed or connected to the lower member **300** at an angle of

approximately 100 degrees to approximately 150 degrees. The lower member **300** includes the front portion **320** and a rear portion **360**. The front axle assembly **23** of the second truck **20** is positioned or mounted on a top surface **332** of the lower member **300**.

(42) The lower member **300** further includes a front opening **336** and a rear opening **366**. The front opening **336** is positioned below the middle member **200**. A front end **401** of the beam member **400** is inserted into the rear opening **366** of the lower member **300**. The beam member **400** may be inserted nearly or all the way through an interior **370** of the lower member **300**. The front end **401** of the beam member **400** may slide or pass through the front opening **336**, such that a substantial portion of the beam member **400** is positioned beneath the upper member **100**.

(43) The front opening **336** is provided with a reinforcing collar **338** that strengthens the front opening **336**. Likewise, the rear opening **366** is strengthened with a reinforcing collar **368**. The reinforcing collars **338** and **368** may circumscribe or nearly fully surround the front opening **336** and the rear opening **366**, respectively

(44) The lower member **300** further includes a side surface **334**. The side surface **334** defines pin openings **394**. The pin openings **394** provide for a pin **470** to pass through the lower member **300** of the towing system **40** and through the beam member **400** in order to lockingly fix the beam member **400** in position in the interior **370** of the lower member **300**. The lower member **300** further includes vertical supports **396** that further strengthen the lower member **300**. The vertical supports **396** are welded to a top widened support plate **380** and to a bottom widened support plate **390**.

(45) The side surface **334** further includes chain mounting members **392** to receive a transport chain **395** or other hoisting or lifting apparatus, which may be used to lift or hoist the assembly **50**. The chain mounting members **392** provide a connection point for chains or other hoisting implements in order for an operator to attach or connect to the towing system **40** and lift the towing system **40**.

(46) With reference to FIG. 2, the top widened support plate **380** is shown. The top widened support plate **380** includes an axle stop **383** and holes **386**. In the embodiments shown in the FIGURES, U clamps **389** pass over the front axle assembly **23** and insert through the holes **386** and are fastened in place with locking nuts, locking pins, etc. The top widened support plate **380** provides a broad surface for the front axle assembly **23** to rest upon. As the top widened support plate **380** widens as it approaches the axle stop **383**, a broad base is provided for the front axle assembly **23** of the second truck **20** that reduces rocking or shaking of the second truck **20** when it is positioned on the top widened support plate **380** of the lower member **300**.

(47) A lower surface of the top widened support plate **380** is welded to the vertical supports **396**, while an upper surface of the bottom widened support plate **390** is welded to the vertical supports **396**. The vertical supports **396** are further welded to the side surfaces **334** of the lower member **300**. In other embodiments, the front axle assembly **23** may be positioned on the top widened support plate **380** and fastened with chains, bolts, fasteners, or other conventional mechanical apparatus that may be used to attach the front axle assembly **23** to the lower member **300**.

(48) The beam member **400** will now be described with reference to FIGS. 2 and 5. The beam member **400** includes the front end **401** and the rear end **450**. As shown in FIG. 5, pin openings **430** are positioned on an exterior of the beam member **400** and travel through the interior width of the beam member **400**. The beam member **400** may include an I-beam construction with an upper surface **422** and a lower surface **424**. A vertical support plate **420** further supports the upper surface **422** and the lower surface **424** of the beam member **400**. The vertical support plate **420** reinforces and strengthens the beam member **400**.

(49) A crossbar **480** assists in connecting a rear portion of the second truck **20** to the assembly **50**. The rear end **450** of the beam member **400** includes a chain receiving member **460** that is fixedly attached to the crossbar **480**. A chain **490** connects lateral portions **482** of the crossbar **480** to the chain receiving member **460**. The crossbar **480** rests on a rear frame **26** of the second truck **20**.

Specifically, a bottom surface **484** of the crossbar **480** rests over the top of the rear frame **26**, while the chain **490** holds the beam member **400** to the crossbar **480**. The chain receiving member **460** may be attached or integral in a fixed engagement to the rear end **450** of the beam member **400**. The chain receiving member **460** includes the chain openings **465** that receive the chain **490**. In other embodiments, the chain receiving member **460** may include other mechanical structure or components to affix to the chain **490**. The length of the chain **490** may be adjusted to raise or lower the rear of second truck **20**. Generally, the front and rear sets of the rear wheels tandem **29** of the second truck **20** should be in contact with the ground to provide better control and braking of the second truck **20**. The length of chain **490** between the crossbar **480** and the chain receiving member **460** may be lengthened to lower the front set of the rear wheels tandem **29**. Likewise, the length of chain **490** between the crossbar **480** and the chain receiving member **460** may be shortened to raise the front set of the rear wheels tandem **29**.

(50) With reference to FIG. 5, the beam member **400** is adjustable in length relative to the lower member **300**. The overall length of the towing system **40** may be adjusted by extending and withdrawing the beam member **400** relative to the lower member **300**. The beam member **400** slides in and out of the interior **370** of the lower member **300**. The upper surface **422** and the lower surface **424** of the beam member **400** may slide against the interior surfaces of the lower member **300**. The circumference or outer surfaces of the beam member **400** should be just smaller than the interior circumference or inner surface of the lower member **300** such that the beam member **400** is held in a secure engagement. As shown in FIG. 9, the lower member **300** and beam member **400** may have box or square cross-sections that provide support and strength to the assembly **40**. The operator may select one of the one or more sets of the pin openings **394** that are spaced along the length of the lower member **300**. The operator may further choose one of the one or more sets of the pin openings **430** along the length of the beam member **400**. This allows a range of length of the towing system **40** that may be used to accommodate different size or types of semi-trucks.

(51) The beam member **400** may be withdrawn through the interior **370** of the lower member **300** and through the front opening **336** such that much of the beam member **400** is directly underneath the upper member **100**. This allows the towing system **40** to collapse into a more convenient storage size. The towing system **40** may further lock in the collapsed configuration.

(52) The semi-trucks that may be towed with the towing saddle system **40** may weigh approximately 5,000 to approximately 10,000 pounds. The towing system **40** may weigh approximately 1,600 pounds. The beam member **400** includes at least one set of the pin openings **430** and may include up to five or six sets of the pin openings **430**. The lower member **300** includes at least one set of the pin openings **394** and up to approximately five or six sets of the pin openings **394**.

(53) The towing system **40**, when using an extendable version of the beam member **400**, provides a range of extension between 0 and approximately 3 linear feet to accommodate a variety of different semi-trucks. When the towing system **40** is in the fully extended configuration, it has an overall length of approximately 18 feet. When the towing system **40** is in the collapsed configuration, it has an overall length of approximately 10 feet. Of course, the kingpin connection assembly **500** and/or the frame connection assembly **600** may be used with other towing systems that do not have an extendable beam member. Thus, the kingpin connection assembly **500** and/or the frame connection assembly **600** may be used with both extendably engaged beam members and with fixedly engaged beam members. For example, the kingpin connection assembly **500** and/or the frame connection assembly **600** may be used with towing systems that have a generally fixed or non-moveable beam or other lower portion that engages a rear of the towed truck. For example, the fixed beam may be welded, bolted, otherwise generally permanently engaged to the remainder of the towing system. For example, the kingpin connection assembly **500** and/or the frame connection assembly **600** may be used with towing systems having no beam member.

(54) The method of operation of the towing system **40** will now be described. First, the towing

system **40** is laid flat on the ground. The beam member **400** is extended or retracted relative to the lower member **300** until it is at the proper length for the second truck **20**. The pin **470** is then inserted through the pin openings **394** in the lower member **300** and through the pin openings **430** of the beam member **400**. The pin **470** is locked in place via a cotter pin, nut or the like. Next, the second truck **20** is driven over the lower member **300** and the beam member **400**. Next, the U clamps **389** are positioned over the front axle assembly **23** and tightened such that the front axle assembly **23** is clamped or fixed to the lower member **300**. Next, the crossbar **480** is passed over the rear frame **26** of the second truck **20**, and the chain **490** connects the crossbar **480** with the chain receiving member **460**.

(55) In one aspect, a crane, hoist, or other lifting implement next lifts the upper member **100** to a sufficient height to where the first truck **10** may back or move underneath the saddle mount portion **124**. Next, the upper member **100** is lowered until the hitch tube **630** is positioned in the saddle mount **15** of the first truck **10**. The hitch tube **630** is then locked to the saddle mount **15**. As such, the towing system **40** is now connected to the first truck **10**, and the second truck **20** is supported via the towing system **40**. The rear wheels tandem **29** of the second truck **20** are in contact with the ground, while the front wheels **27** of the second truck **20** are approximately 6" to approximately 12" off and above the ground. In other aspects, as further described below, the beam member **400** may include a first transport opening **402** and a second transport opening **404** that are complementary or matching to a third transport opening **306** in the lower member **300**. The transport openings **402**, **404**, and **306** provide for access entry points for a forklift, skid-steer, or other moving apparatus with a fork or arm members to engage with the towing system **40** in order to lift and move the towing system **40**.

(56) With reference to FIG. 5, the towing system **40** collapses to a collapsed configuration for efficient storage and convenient maneuverability. The beam member **400** is withdrawn or retracted through the interior **370** of the lower member **300** and through the front opening **336** such that much of the beam member **400** is directly underneath the upper member **100**. The beam member **400** may be locked in place with the pin **470**. In this collapsed configuration, a center of gravity of the towing system **40** is positioned approximately in the middle of the collapsed towing system **40**. The chain mounting members **392** provide the connection point for chains or other hoisting implements in order for an operator to attach or connect to the towing system **40** and lift the towing system **40**. Thus, the towing system **40** may be safely lifted at its middle point without the towing system **40** twisting or torquing on the transport chain **395**. Due to its compact size, up to approximately 25 of the towing systems **40** may be stored on a standard flatbed trailer. Of course, the transport openings **402**, **404**, and **306** also provide for access entry points for a forklift, skid-steer, etc. lift and move the towing system **40**.

(57) The towing system **40** is convertible for use between towing trucks with a saddle mounts, fifth wheel mounts, or other hitches and mounts. The towing system **40** may include one or more connection assemblies that are configured to engage with the saddle mounts, fifth wheel mounts, or other hitches and mounts of the towing truck. For example, a first connection assembly may be removed from the towing system **40** and replaced with a second connection assembly. The one or more connection assemblies are configured to operate with a specific mount of the towing truck, such as saddle mounts, fifth wheel mounts, or other hitches and mounts. Thus, a single towing system **40** may be provided with different connection assemblies to match the mount of the towing truck. A front portion of the connection assemblies removably engages or connects with the first truck **10**. A rear portion of the connection assemblies removably engages or connects with the upper member **100** of the towing system **40**.

(58) With respect to FIG. 1, the towing system **40** includes the upper member **100** with the front portion **118** that is configured to removably connect or engage to the saddle mount **15** of the first truck **10** or to a fifth wheel mount (not shown) of the first truck **10** or to the saddle mount or to the fifth wheel mount of a different truck. In this aspect, the front portion **118** engages with or receives

either the kingpin connection assembly **500** or the frame connection assembly **600**. Of course, other connection assemblies may be used with upper member **100**. The kingpin connection assembly **500** is used when the first truck **10** (or other towing truck) has the fifth wheel mount installed. The frame connection assembly **600** is used when the first truck **10** (or other towing truck) has the saddle mount **15** installed. The kingpin connection assembly **500** and the frame connection assembly **600** may be used with the upper member **100**. With respect to FIGS. 2-5, the upper member **100** is shown with the frame connection assembly **600**.

(59) The kingpin connection assembly **500** and the frame connection assembly **600** are removably connected or engaged to the upper member **100** of the towing system **40**. The kingpin connection assembly **500** and the frame connection assembly **600** are interchangeable depending upon the preferred or available mount on the towing truck. The same towing system **40** may use either the kingpin connection assembly **500** or the frame connection assembly **600** depending on whether the towing truck has a fifth wheel mount or saddle mount. As such, the towing system **40** may be easily converted to connect to either a fifth wheel mount or saddle mount. These features improve efficiency, reduce costs, and save time. Further, the towing system **40** may be repeatedly converted or changed for use with either the fifth wheel mount or the saddle mount. The kingpin connection assembly **500** and the frame connection assembly **600** may be readily installed by sliding either the kingpin connection assembly **500** or the frame connection assembly **600** into the upper member **100**.

(60) In this aspect, the front portion **118** of the upper member **100** of the towing system **40** is generally hollow and serially receives either the kingpin connection assembly **500** or the frame connection assembly **600**. Thus, the towing system **40** is convertible for use with two different styles of mounting, i.e., the saddle mount **15** or the fifth wheel mount.

(61) The kingpin connection assembly **500** will now be described with reference to FIGS. 17-23. The kingpin connection assembly **500** includes a tube **510**. The tube **510** is a generally linear member. A coupler plate **530** is mounted to a front portion **511** of the tube **510**. In this aspect, the coupler plate **530** is mounted below a lower surface **513** of the tube **510** at the front portion **511** of the tube **510**. In this aspect, mounting gussets **540** are welded to lateral sides **514** of the tube **510** and to the coupler plate **530**. The tube **510** slides into a generally hollow interior **121** of the front portion **118** of the upper member **100**. In this aspect, a rear portion **518** of the tube **510** slides into the generally hollow interior **121** of the front portion **118** of the upper member **100**. The rear portion **518** of the tube **510** has an outer diameter less than an inner diameter of the front portion **118** of the upper member **100**. The tube **510** may include a larger outer diameter closer to the front portion **511** of the tube **510**.

(62) The coupler plate **530** includes a generally circular outer diameter **532**. A kingpin **550** is mounted to the coupler plate **530**. The kingpin **550** lockingly engages with the fifth wheel mount. With respect to FIGS. 21 and 22, in this aspect, the kingpin **550** includes a mounting plate portion **552** and the kingpin portion **556**. The kingpin portion **556** passes through a central circular opening **533** of the coupler plate **530**, and the mounting plate portion **552** is welded or otherwise fastened to the coupler plate **530**, with the kingpin portion **556** pointing downward. The mounting plate portion **552** has an outer diameter **558** generally larger than the central circular opening **533** of the coupler plate **530**. The lower surfaces **542** of the mounting gussets **540** are welded to an upper surface **536** of the coupler plate **530**. In this aspect, the lower surfaces **542** of the mounting gussets **540** are welded to the upper surface **536** of the coupler plate **530** just outside of the mounting plate portion **552**. The lower surfaces **542** of the mounting gussets **540** may be shaped or configured to not contact the mounting plate portion **552** of the kingpin **550**.

(63) The rear portion **518** of the tube **510** includes one or more sets of openings **560** passing through the lateral sides **514** of the tube **510**. Similarly, the front portion **118** of the upper member **100** includes one or more sets of openings **128** passing through lateral sides **126** of the front portion **118**. Removable fasteners, such as bolts, pins, rods, etc. pass through the openings **128** and the

openings **560** and are fastened in place. In the aspect shown in the FIGS, pin sleeves **570** are aligned with the openings **560**. Linch pins **580** pass through the openings **560**, the openings **128**, and an inner diameter of the pin sleeves **570** to affix the kingpin connection assembly **500** to the front portion **118** of the upper member **100**. The linch pins **580** may receive a cotter pin assembly or other lock to hold the linch pins **580** in place.

(64) The kingpin connection assembly **500** may be formed from solid, hollow or partially hollow sections of square tubing, round tubing, or stock. If the kingpin connection assembly **500** is formed from solid carbon steel, then the one or more sets of openings **560** are bored through the width of the tube **510**. The kingpin connection assembly **500** may be made from low to medium strength carbon steel.

(65) With respect to FIG. **17**, an upper shim plate **590** is mounted to an upper surface **512** of the tube **510** spaced a distance from a rear end **519** of the rear portion **518** of the tube **510**. A lower shim plate **595** is mounted to the lower surface **513** of the tube **510** proximate the rear end **519** of the rear portion **518** of the tube **510**. The upper and lower shim plates **590** and **595** take up space in the generally hollow interior **121** of the upper member **100** to make a tighter fit between the tube **510** and the generally hollow interior **121** of the front portion **118** of the upper member **100**.

(66) The tube **510** may also include a more sets of openings **129** passing through the lateral sides **514** of the tube **510**. These openings **129** are used to lock the tube **510** in place in a collapsed or storage configuration. When the towing system **40** is not in use, for example on a return trip after a delivery, the tube **510** may be withdrawn into the front portion **118** of the upper member **100** past the openings **560** and locked in place by inserting removable fasteners, such as a bolt, pin, rod, lock, etc. through the openings **129**. This locks the towing system **40** into the collapsed or storage configuration.

(67) The frame connection assembly **600** will now be described with reference to FIGS. **11-16**. The frame connection assembly **600** includes a tube **610**. The tube **610** is a generally linear member. A hitch tube **630** is mounted below a lower surface **612** of the tube **610** at a front portion **622** of the tube **610**. The saddle mount **15** lockingly engages with the hitch tube **630**.

(68) The hitch tube **630** is mounted generally perpendicular to length of the tube **610**. An upper surface **632** of the hitch tube **630** may be directly welded or otherwise fastened to the lower surface **612** of the tube **610**. In this aspect, mounting gussets **640** are welded to the lower surface **612** of the tube **610** and to a rear surface **634** of the hitch tube **630**. The mounting gussets **640** may include a generally triangular shape with a first side **645** welded to the lower surface **612** of the tube **610** and a second side **648** welded to the rear surface **634** of the hitch tube **630**, wherein the first side **645** and the second side **648** are adjacent sides of the generally triangular shape.

(69) The tube **610** slides into a generally hollow interior **121** of the front portion **118** of the upper member **100**. In this aspect, a rear portion **618** of the tube **610** slides into the generally hollow interior **121** of the front portion **118** of the upper member **100**. The rear portion **618** of the tube **610** has an outer diameter less than an inner diameter of the front portion **118** of the upper member **100**. The tube **610** may include a larger outer diameter closer to the front portion **622** of the tube **610**. The hitch tube **630** includes a generally square or rectangular outer surface. Hitch rods **641**, **642** and **644** are mounted to the hitch tube **630**. In this aspect, a first hitch rod **641** is mounted parallel to the rear surface **634** of the hitch tube **630**, a second hitch rod **642** is mounted parallel to the front surface **636** of the hitch tube **630**, and a third hitch rod **644** is mounted parallel to the front surface **636** of the hitch tube **630** above the second hitch rod **642**.

(70) The rear portion **618** of the tube **610** includes one or more sets of openings **660** passing through the lateral sides **614** of the tube **610**. Similarly, the front portion **118** of the upper member **100** includes the one or more sets of openings **128** passing through the lateral sides **126** of the front portion **118**. Removable fasteners, such as bolts, pins, rods, etc. pass through the openings **128** and the openings **660** and are fastened in place. In the aspect shown in the FIGS, pin sleeves **670** are aligned with the openings **660**. Linch pins **680** pass through the openings **128**, the openings **660**,

and an inner diameter of the pin sleeves **670** to affix the frame connection assembly **600** to the front portion **118** of the upper member **100**. The linch pins **680** may receive a cotter pin assembly or other locking device to hold the linch pins **680** in place. The frame connection assembly **600** may be formed from solid, hollow or partially hollow sections of square tubing, round tubing, or stock. If the frame connection assembly **600** is formed from solid carbon steel, then the one or more sets of openings **660** are bored through the width of the tube **610**. The frame connection assembly **600** may be made from low to medium strength carbon steel. With respect to FIGS. **11** and **12**, an upper shim plate **690** is mounted to an upper surface **632** of the tube **610** spaced a distance from a rear end of the rear portion **618** of the tube **610**. A lower shim plate **695** is mounted to the lower surface **612** of the tube **610** proximate the rear end of the rear portion **618** of the tube **610**. The upper and lower shim plates **690** and **695** take up space in the generally hollow interior **121** of the upper member **100** to make a tighter fit between the tube **610** and the generally hollow interior **121** of the front portion **118** of the upper member **100**.

(71) In other aspects, the kingpin connection assembly **500** and the frame connection assembly **600** are removably connected or engaged to an upper member of a towing system having a beam member that is fixedly engaged to a lower member or having a beam member **400** that is integral with the lower member—such that the beam member **400** is stationary relative to the lower member. In other aspects, the kingpin connection assembly **500** and the frame connection assembly **600** are removably connected or engaged to an upper member of a towing system having a lower member that supports the front axle assembly **23** of the second truck and the rear frame **26** of the second truck **20**.

(72) In another aspect, as shown in FIGS. **27-31**, the beam member **400** includes the first transport opening **402** and the second transport opening **404** that are complementary or matching to the third transport opening **306** in the lower member **300**. The transport openings **402**, **404**, and **306** provide for access entry points for a forklift, skid-steer, or other moving apparatus with a fork or arm members to engage with the towing system **40** in order to lift and move the towing system **40**.

(73) As described above, the beam member **400** is withdrawn or retracted through the interior **370** of the lower member **300** and through the front opening **336** such that much of the beam member **400** is directly underneath the upper member **100** and the first transport opening **402**. The beam member **400** is withdrawn or retracted sufficiently through the interior **370** of the lower member **300** of the beam member **400** such that the first transport opening **402** is aligned with the third transport opening **306** in the lower member **300**. The beam member **400** may be locked in place with the pin **470**. In moving the beam member **400** to the collapsed configuration, the beam member **400** is withdrawn or retracted through the interior **370** of the lower member **300** sufficiently such that the first transport opening **402** and the third transport opening **306** are in alignment, while the second transport opening **404** is outside of the lower member **300**. Now, the moving apparatus may lift or move the entire towing system **40** in a single operation. A first fork or arm of the moving apparatus may pass through the combination of the first transport opening **402** and the third transport opening **306** while a second fork or arm of the same moving apparatus may pass the second transport opening **404** in order lift or move the towing system **40**, which may weigh approximately 1600 pounds. Thus, the first and second forks or arms may generally simultaneously pass through the three transport openings **402**, **404**, and **306**.

(74) The first transport opening **402** and the second transport opening **404** are both in a side of the beam member **400**, and a space or distance between the first transport opening **402** and the second transport opening **404** is constant and is determined by the spacing between the forks or arms of the moving apparatus. In other aspects, the relative positions or spacing of the three transport openings **402**, **404**, and **306** may be adjusted to accommodate different types and styles of moving apparatus. The relative positions or spacing of the three transport openings **402**, **404**, and **306** may also be adjusted to accommodate different sizes and weights of towing systems **40**. The first and second transport openings **402**, **404** of the beam member **400** are positioned closer to the front end **401** of

the beam member **400**.

(75) The first transport opening **402** and the second transport opening **404** pass entirely through a side of the beam member **400**. Similarly, the third transport opening **306** passes entirely through a side of the lower member **300**. The third transport opening **306** in the lower member **300** is positioned proximate the junction of the lower member **300** and the middle member **200**. The third transport opening **306** in the lower member **300** is positioned proximate the front opening **336** in the lower member **300**. The three transport openings **402**, **404**, and **306** may include rectangular shaped openings with reinforcement.

(76) In other aspects, the kingpin connection assembly **500** and the frame connection assembly **600** may be used with other towing assemblies. The use of the kingpin connection assembly **500** and the frame connection assembly **600** are not limited to towing systems **40** with a collapsible beam member **400**. For example, the kingpin connection assembly **500** and the frame connection assembly **600** may be used with towing assemblies that have a fixed version of the beam member **400**, a one-piece version of the lower member **300** that lacks a beam member **400**, a rigid combination of the lower member **300** and the beam member **400**, etc.

(77) It should be understood from the foregoing that, while particular embodiments of the invention have been illustrated and described, various modifications can be made thereto without departing from the spirit and scope of the present invention. Therefore, it is not intended that the invention be limited by the specification; instead, the scope of the present invention is intended to be limited only by the appended claims.

Claims

1. A truck towing system, comprising: an upper member; a connection assembly comprising a tube, a front portion of the tube configured to removably engage or connect with a first truck; a rear portion of the tube configured to removably engage or connect with the upper member; a lower member, wherein the lower member joins the upper member in a fixed engagement; the lower member supports a front portion of a second truck; a beam member, the beam member engaged to the lower member; the beam member configured to connect to a rear portion of the second truck; wherein the connection assembly is a kingpin connection assembly, a coupler plate is mounted to a front portion of the tube, and a kingpin is mounted to the coupler plate; and wherein a rear portion of the tube is configured to removably slide into a generally hollow interior of a front portion of the upper member.

2. The truck towing system according to claim 1, wherein the tube is linearly shaped.

3. The truck towing system according to claim 1, wherein the connection assembly includes a frame connection assembly and the kingpin connection assembly, and one of the frame connection assembly and the kingpin connection assembly are removably engaged or connected with the upper member.

4. The truck towing system according to claim 1, wherein the connection assembly is configured to be replaced with a different connection assembly.

5. The truck towing system according to claim 1, wherein the rear portion of the tube has an outer diameter less than an inner diameter of the front portion of the upper member.

6. The truck towing system according to claim 1, wherein the kingpin includes a mounting plate portion and a kingpin portion, the kingpin portion passes through a central circular opening of the coupler plate, and the mounting plate portion is welded or otherwise fastened to the coupler plate with the kingpin portion pointing downward.

7. The truck towing system according to claim 1, wherein the rear portion of the tube includes one or more sets of openings passing through lateral sides of the tube, a front portion of the upper member includes one or more sets of openings passing through lateral sides of the front portion, and removable fasteners pass through the openings of the lateral sides of the tube and the openings

of the lateral sides of the front portion, and the fasteners hold the rear portion of the tube in the front portion of the upper member.

8. The truck towing system according to claim 1, wherein the beam member slides in and out of an interior of the lower member, the beam member slides to a collapsed configuration, and the beam member slides to an extended configuration.

9. The truck towing system according to claim 1, wherein the truck towing system is adjustable in length by extending or retracting the beam member relative to the lower member, wherein the beam member slides in and out of a rear opening of the lower member.

10. The truck towing system according to claim 1, wherein the beam member includes a first transport opening and a second transport opening, and the lower member includes a third transport opening, and the beam member is withdrawn or retracted such that the first transport opening is aligned with the third transport opening in the lower member.

11. A truck towing system, comprising; an upper member; a connection assembly comprising a tube, a front portion of the tube configured to removably engage or connect with a first truck; a rear portion of the tube configured to removably engage or connect with the upper member; a lower member, wherein the lower member joins the upper member in a fixed engagement; the lower member supports a front portion of a second truck; a beam member, the beam member engaged to the lower member; the beam member configured to connect to a rear portion of the second truck; wherein the connection assembly is a kingpin connection assembly, a coupler plate is mounted to a front portion of the tube, and a kingpin is mounted to the coupler plate; and wherein a rear portion of the tube has an outer diameter less than an inner diameter of a front portion of the upper member.

12. A truck towing system, comprising: an upper member; a frame connection assembly configured to removably engage or connect with a first truck; the frame connection assembly configured to removably engage or connect with the upper member; a lower member, wherein the lower member joins the upper member in a fixed engagement; the lower member supports a front portion of a second truck; a beam member, the beam member engaged to the lower member; the beam member configured to connect to a rear portion of the second truck; the frame connection assembly includes a tube, and a hitch tube is mounted to a lower surface of the tube generally perpendicular to the tube; and wherein mounting gussets are welded to the lower surface of the tube and to a rear surface of the hitch tube, and one or more hitch rods are welded to the lower surface of the tube or to a rear surface of the hitch tube.

13. A truck towing system, comprising: an upper member; a connection assembly comprising a tube, a front portion of the tube configured to removably engage or connect with a first truck; a rear portion of the tube configured to removably engage or connect with the upper member; a lower member, wherein the lower member joins the upper member in a fixed engagement; the lower member supports a front portion of a second truck; a beam member, the beam member engaged to the lower member; the beam member configured to connect to a rear portion of the second truck; wherein the connection assembly is a frame connection assembly and a hitch tube is mounted to a lower surface of the tube generally perpendicular to the tube; wherein the rear portion of the tube has an outer diameter less than an inner diameter of a front portion of the upper member.

14. A truck towing system, comprising: an upper member; a connection assembly comprising a tube, a front portion of the tube configured to removably engage or connect with a first truck; a rear portion of the tube configured to removably engage or connect with the upper member; a lower member, wherein the lower member joins the upper member in a fixed engagement; the lower member supports a front portion of a second truck; a beam member, the beam member engaged to the lower member; the beam member configured to connect to a rear portion of the second truck; wherein the connection assembly is a frame connection assembly and a hitch tube is mounted to a lower surface of the tube generally perpendicular to the tube; wherein the tube of the connection assembly is solid, hollow, or partially hollow; wherein the rear portion of the tube is configured to

removably slide into a generally hollow interior of a front portion of the upper member.

15. The truck towing system according to claim 14, wherein the rear portion of the tube includes one or more sets of openings passing through lateral sides of the tube, the front portion of the upper member includes one or more sets of openings passing through lateral sides of the front portion, and removable fasteners pass through the openings of the lateral sides of the tube and the openings of the lateral sides of the front portion, and the fasteners hold the rear portion of the tube in the front portion of the upper member.

16. A truck towing system for use with a saddle mount or a kingpin mount, comprising: an upper member; a connection assembly configured to removably engage or connect with a towing truck having a saddle mount or a kingpin mount, wherein the connection assembly is a frame connection assembly or a kingpin connection assembly; a rear portion of the connection assembly includes a linear shaped tube that is configured to removably engage or connect with the upper member; a lower member, wherein the lower member joins the upper member in a fixed engagement; the lower member configured to support a front portion of a towed truck; a beam member, the beam member engaged to the lower member; the beam member configured to connect to a rear portion of the towed truck; and wherein the tube is configured to removably slide into an interior of a front portion of the upper member.

17. The truck towing system according to claim 16, wherein the beam member includes a first transport opening and a second transport opening, and the lower member includes a third transport opening, and the beam member is withdrawn or retracted such that the first transport opening is aligned with the third transport opening in the lower member.

18. A method of converting a truck towing system for use with a saddle mount or a kingpin mount, comprising: providing an upper member and a lower member, wherein the lower member joins the upper member in a fixed engagement, the lower member configured to support a front portion of a towed truck; providing a beam member, the beam member engaged to the lower member, and the beam member configured to connect to a rear portion of the towed truck; determining whether a towing truck has a saddle mount or a kingpin mount; providing a frame connection assembly with a first tube and providing a kingpin connection assembly with a second tube; engaging or connecting the frame connection assembly or the kingpin connection assembly with the upper member, the frame connection assembly configured to removably engage or connect with the saddle mount of the towing truck, and the kingpin connection assembly configured to removably engage or connect with the kingpin mount of the towing truck; and sliding a rear portion of the first tube or the second tube into an interior of a front portion of the upper member.

19. The method according to claim 18, further comprising disengaging or disconnecting the frame connection assembly or the kingpin connection assembly from the upper member.

20. A method of using of a truck towing system comprising: providing an upper member and a lower member, wherein the lower member joins the upper member in a fixed engagement, the lower member configured to support a front portion of a towed truck; providing a beam member, the beam member engaged to the lower member, and the beam member configured to connect to a rear portion of the towed truck; providing a frame connection assembly and a kingpin connection assembly; selecting the frame connection assembly or the kingpin connection assembly to provide a selected connection assembly, the frame connection assembly configured to engage a saddle mount of a towing truck and the kingpin connection assembly configured to engage a kingpin mount of the towing truck; engaging the selected connection assembly with the upper member; and inserting a rear portion of the selected connection assembly into an interior of a front portion of the upper member.

21. The method of using of a truck towing system according to claim 20, further comprising fastening the rear portion of the selected connection assembly to the front portion of the upper member.

22. The method of using of a truck towing system according to claim 20, further comprising

removing the selected connection assembly from the upper member, and replacing the selected connection assembly with the other of the frame connection assembly or the kingpin connection assembly.

23. The method of using of a truck towing system according to claim 20, further comprising connecting the rear portion of the towed truck to the beam member and connecting the selected connection assembly to the towing truck.
